

Gradient-Preserving Activation Clipping: MNIST Report

ByzFL SNN Benchmark

July 23, 2026

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1 Activation Definitions

All variants below share the same MNIST CNN architecture ($\text{Conv2d}(1, 20, 5, 1) \rightarrow \text{Conv2d}(20, 50, 5, 1) \rightarrow \text{Linear}(800, 500) \rightarrow \text{Linear}(500, 10)$), differing only in the activation applied after the first three layers (the final layer always uses `log_softmax`). Forward/backward definitions below are transcribed directly from `byzfl/fed_framework/clipped_activations` and `byzfl/fed_framework/client.py`.

1.1 Baseline: Hardtanh Clip (existing, true derivative)

Forward and backward are both the standard `hardtanh` clamp – the gradient genuinely vanishes above the clip value:

$$y = \text{clamp}(x, 0, C)$$

$$\frac{\partial y}{\partial x} = \begin{cases} 1 & 0 < x < C \\ 0 & \text{otherwise} \end{cases}$$

1.2 STE Clip (ClippedReLU_STE)

Forward is the same hard clamp; the backward pass is a straight-through estimator that never zeroes the gradient for positive inputs, regardless of saturation:

$$y = \text{clamp}(x, 0, C)$$

$$\frac{\partial y}{\partial x} = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$

1.3 Ramp Clip (ClippedReLU_Ramp)

Forward is the same hard clamp to $[0, C]$. The backward pass keeps slope 1 up to C , then decays *linearly* to 0 over $[C, rC]$ (ramp ratio r ; $r = 2$ in all runs here, so the ramp ends at $2C$) – this preserves the distinction between a coordinate that is slightly over C and one that is far over C , instead of treating both identically:

$$y = \text{clamp}(x, 0, C)$$

$$\frac{\partial y}{\partial x} = \begin{cases} 1 & 0 < x \leq C \\ \frac{rC - x}{rC - C} & C < x < rC \\ 0 & x \leq 0 \text{ or } x \geq rC \end{cases}$$

1.4 Fixed Gradient-Norm Clip (gradient_clip_val= 21)

Also not an activation function – a plain-ReLU CNN where each honest client’s overall gradient (all parameters concatenated) is rescaled, via `torch.nn.utils.clip_grad_norm_`, to a fixed global L2 norm threshold $C_g = 21$, applied to the raw per-parameter `.grad` tensors right after `loss.backward()` – i.e. before flattening and before the momentum accumulation described below:

$$g = \nabla_{\theta} \mathcal{L}(\theta) \quad (\text{concatenation of all parameter gradients})$$

$$g^{\text{clipped}} = \begin{cases} g \cdot \frac{C_g}{\|g\|_2} & \|g\|_2 > C_g \\ g & \text{otherwise} \end{cases}$$

Unlike the adaptive client-side clip below, C_g is a fixed constant (not data-adaptive) and is not a quantile of anything.

1.5 Adaptive Per-Coordinate Quantile Clip (AdaptiveQuantileClip)

Instead of a fixed constant C , the clip threshold is recomputed every forward pass as the τ -quantile of that layer’s own activation coordinates (detached, i.e. treated as a constant for backpropagation), so the threshold adapts to the current activation distribution instead of being hand-picked. Two backward variants were implemented – *plain* (same failure mode as `hardtanh`: hard zero above C) and *STE* (gradient survives above C):

$$C = \text{quantile}_{\tau}(\text{flatten}(x)) \quad (\text{detached})$$

$$y = \text{clamp}(x, 0, C)$$

$$\frac{\partial y}{\partial x} = \begin{cases} \mathbb{I}[0 < x \leq C] & \text{plain} \\ \mathbb{I}[x > 0] & \text{STE} \end{cases}$$

Swept at $\tau \in \{0.80, 0.90\}$, both backward variants (*results pending – not yet available at time of writing*).

1.6 Adaptive Client-Side Gradient-Norm Clip

Not an activation function – a plain-ReLU CNN where each honest client clips the L2 norm of the (post-momentum) gradient vector it sends to the server, to the τ -quantile of its own last $W = 100$ gradient norms (a sliding window, since gradient norms are non-stationary and shrink over training). Let v_t be the momentum gradient at step t and $n_t = \|v_t\|_2$; a history buffer of the last W values of n_t is maintained *before* any clipping is applied, so the quantile reflects the true, unclipped distribution:

$$\begin{aligned}\tau\text{-threshold}_t &= \text{quantile}_\tau(\{n_{t-W+1}, \dots, n_t\}) \\ v_t^{\text{sent}} &= \begin{cases} v_t \cdot \frac{\tau\text{-threshold}_t}{n_t} & n_t > \tau\text{-threshold}_t \\ v_t & \text{otherwise} \end{cases}\end{aligned}$$

The internal momentum buffer itself is never rescaled – only the copy actually sent to the server is clipped. Swept at $\tau \in \{0.70, 0.80\}$ (*results pending – not yet available at time of writing*).

2 Best Overall Performance (Worst-Case Across Attacks)

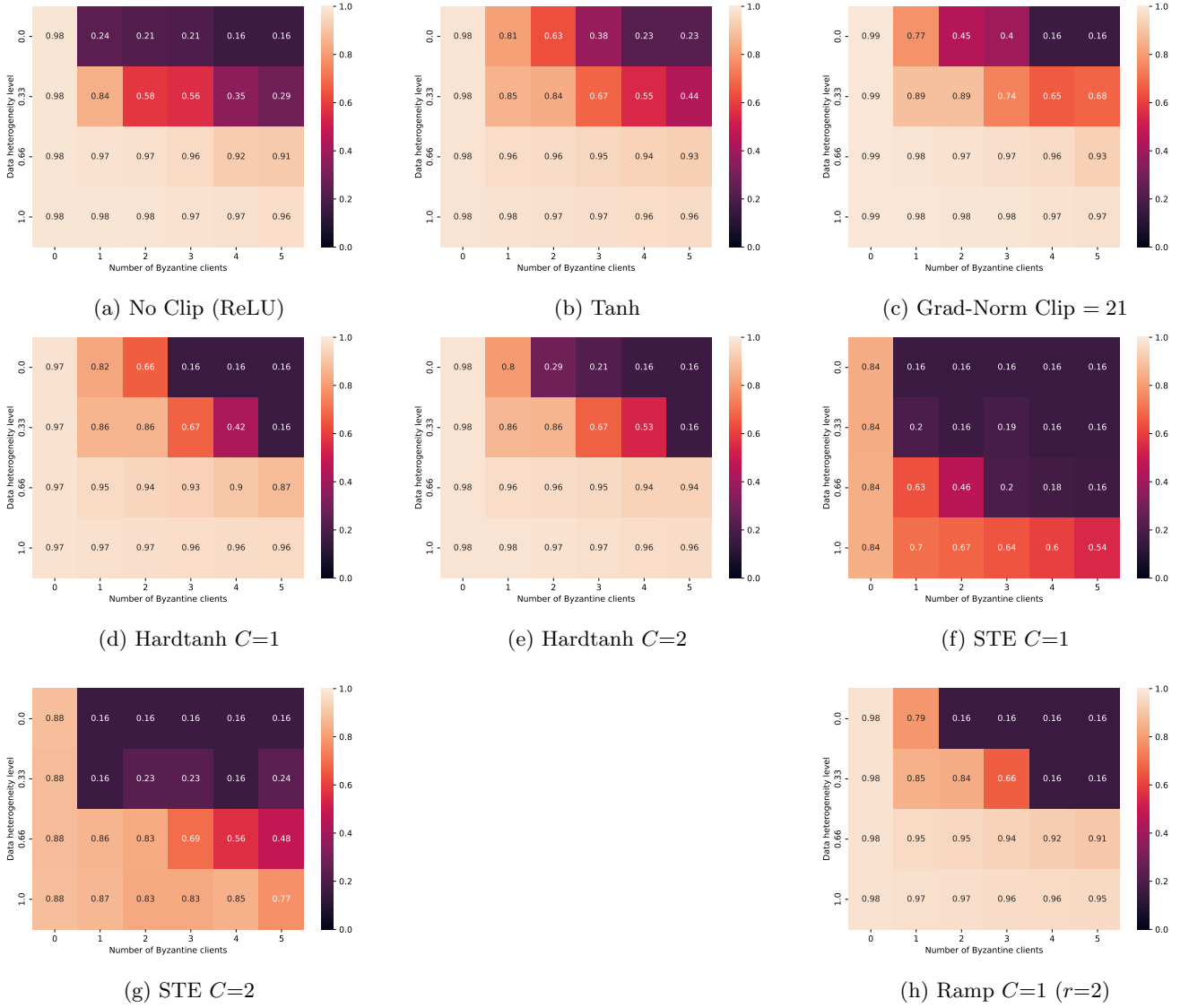


Figure 1: Best overall test accuracy (worst-case across attacks).

3 Best Aggregator under Specific Attacks

3.1 Optimal ALIE Attack

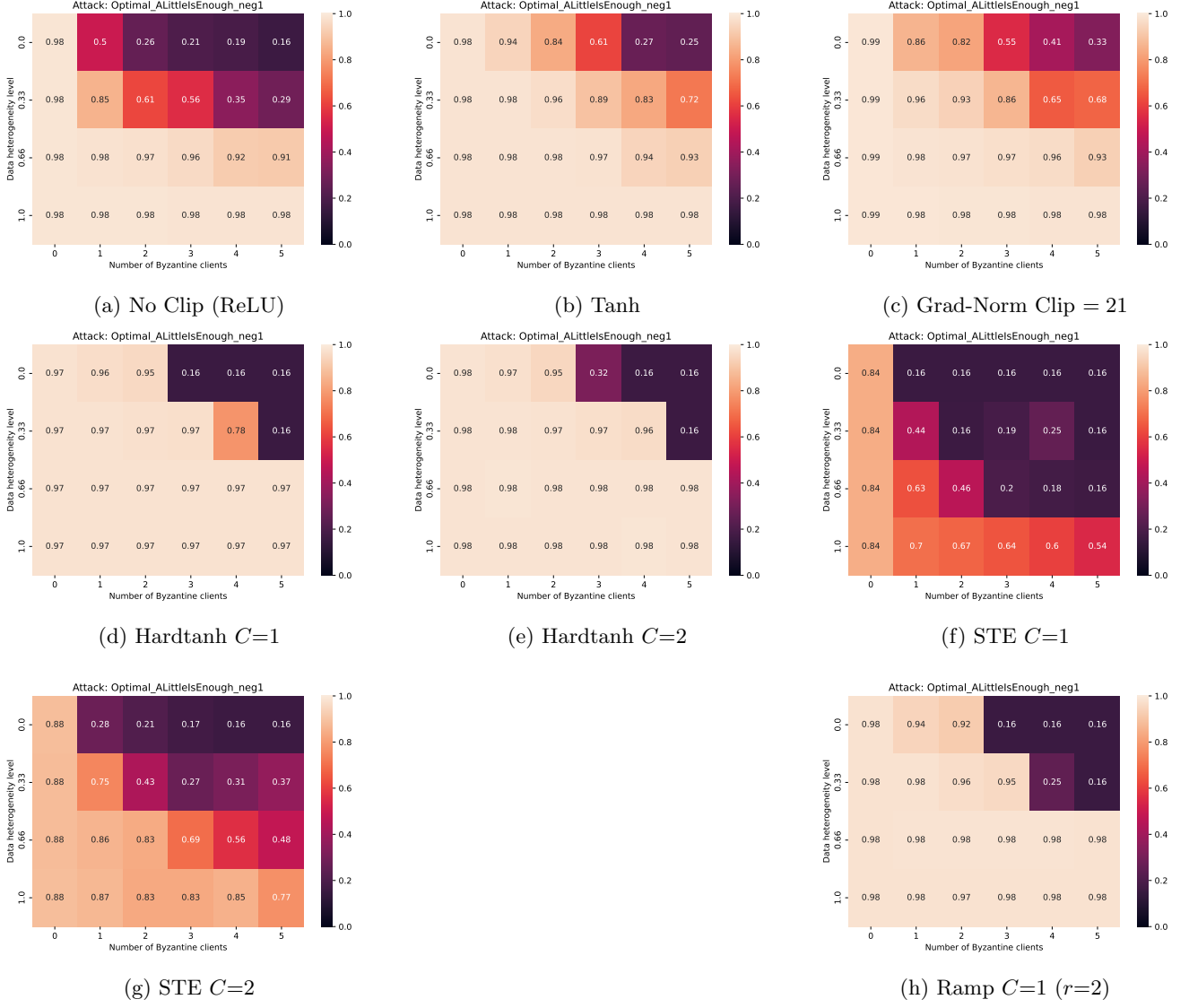


Figure 2: Best test accuracy under ALIE.

3.2 Sign Flipping Attack

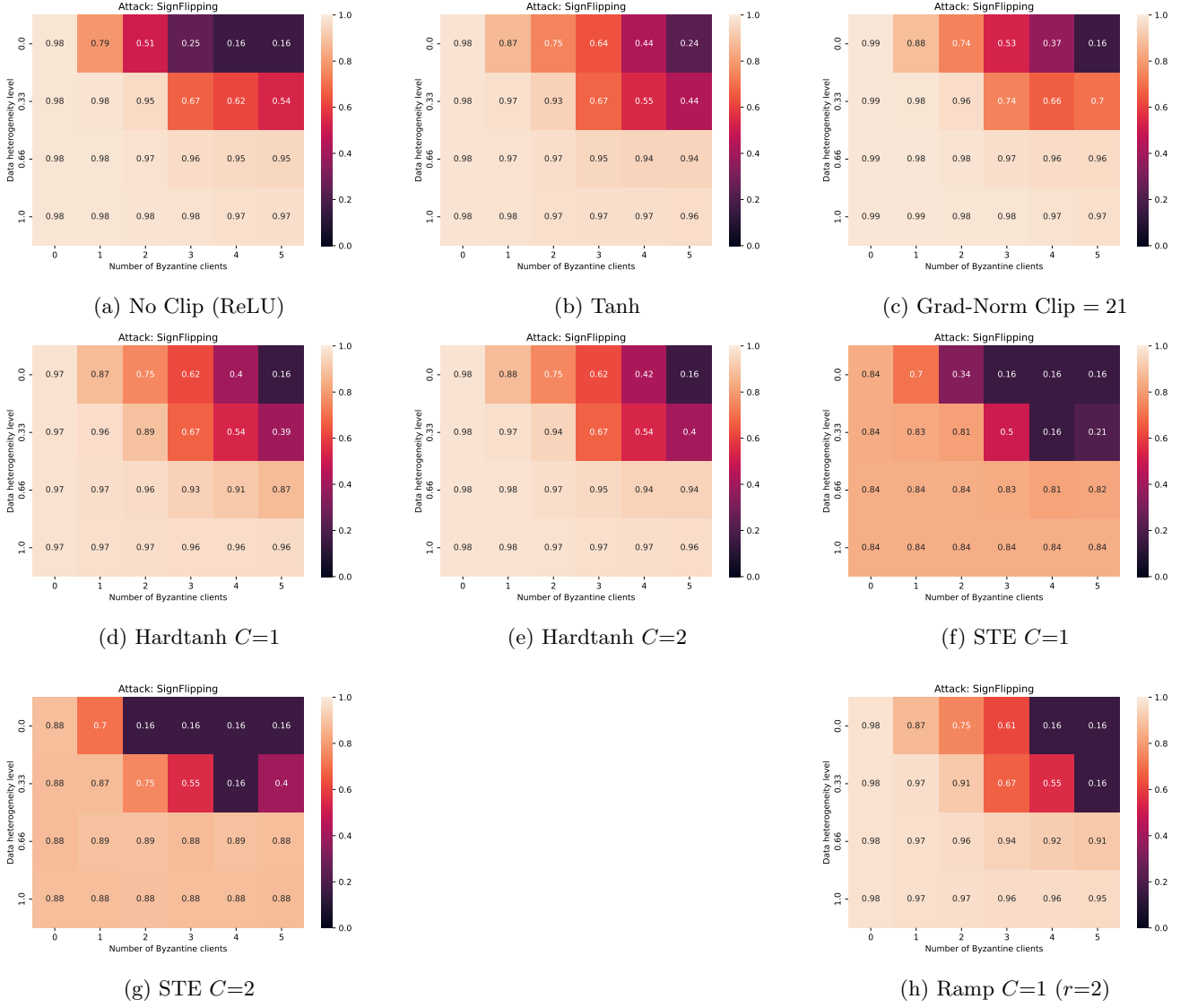


Figure 3: Best test accuracy under SF.

3.3 Optimal IPM Attack

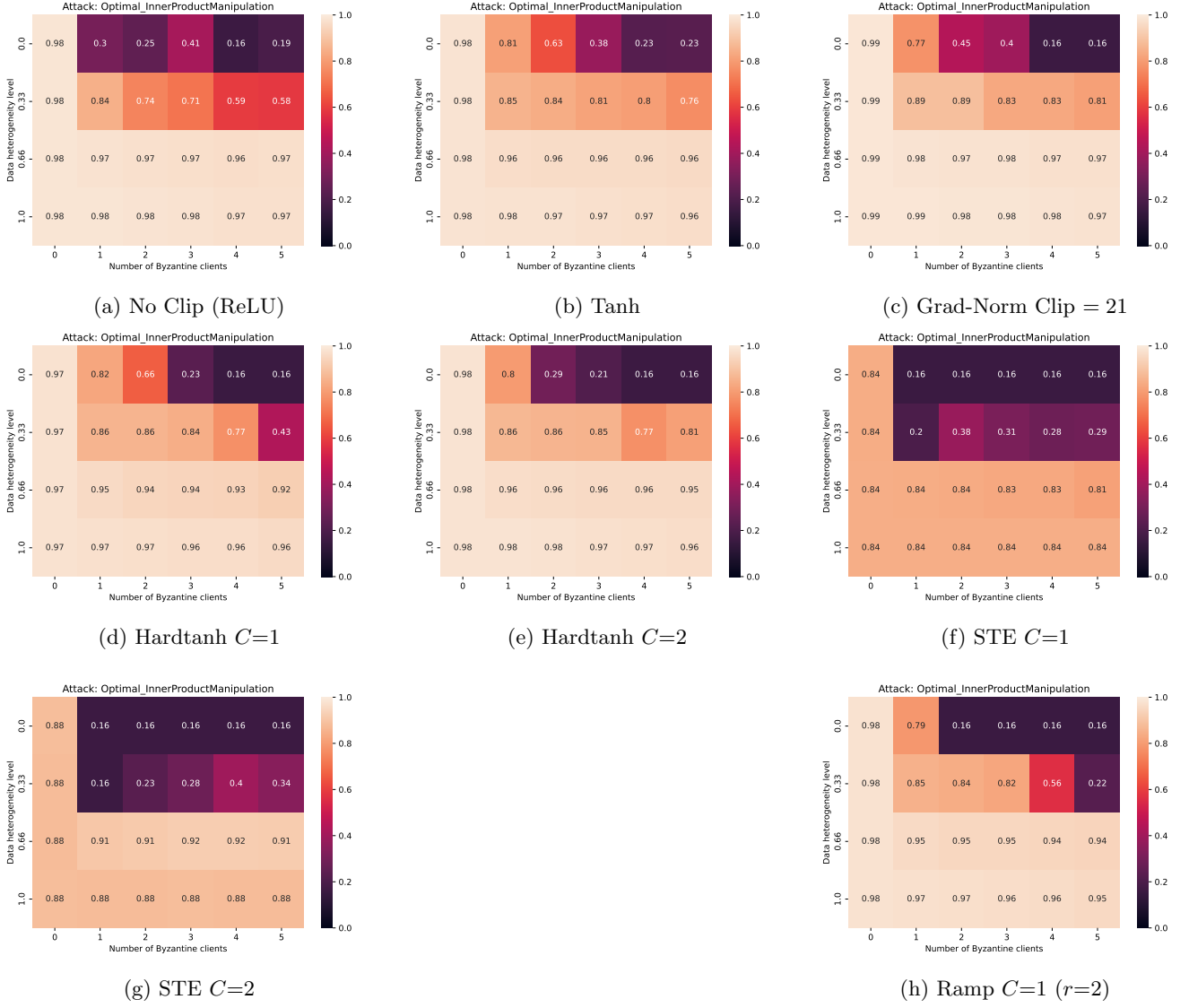


Figure 4: Best test accuracy under IPM.

4 Centered Clipping (CC)

4.1 CC under Worst-Case across all Attacks

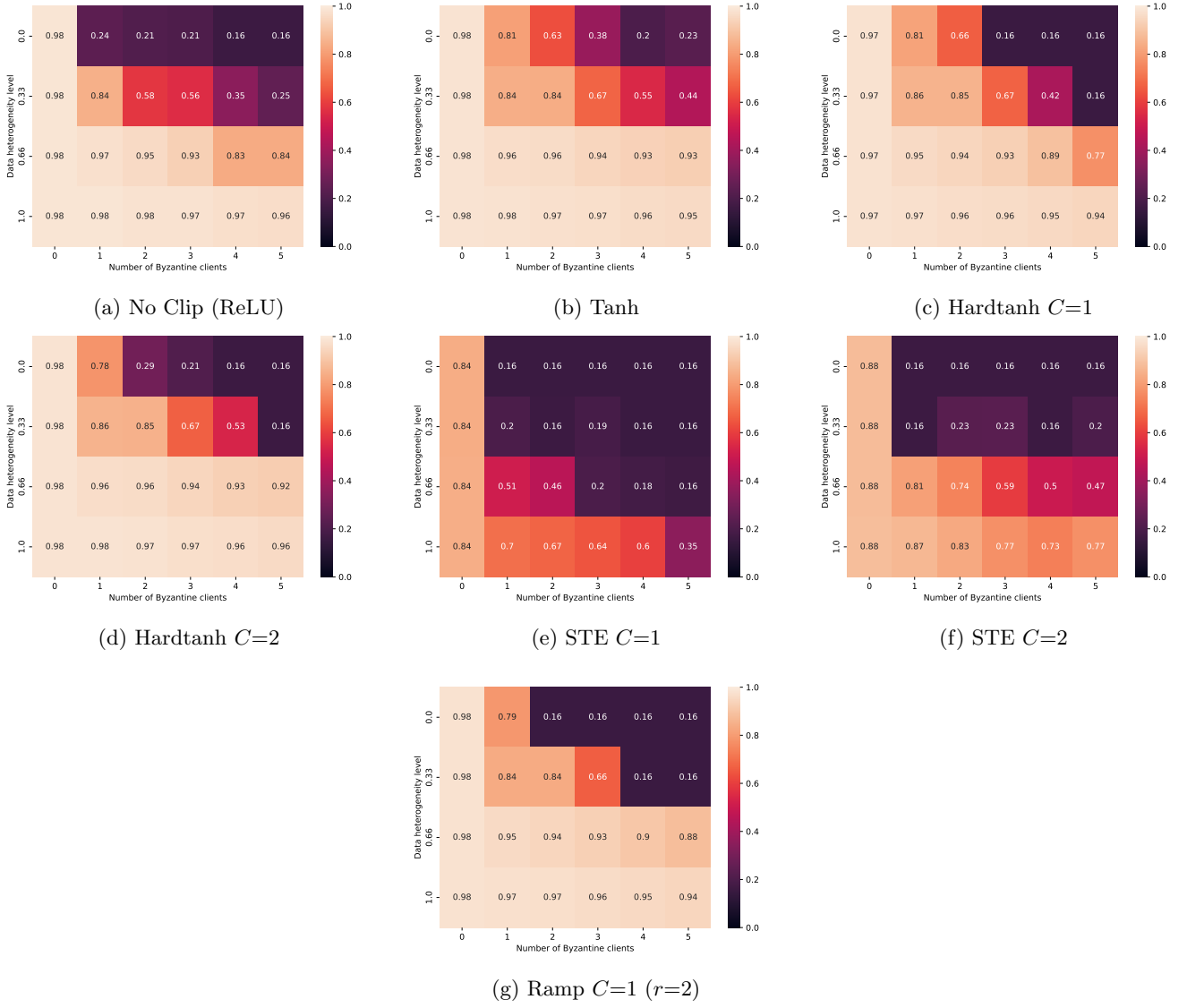


Figure 5: Centered Clipping (CC) under Worst-Case across all Attacks.

4.2 CC under ALIE

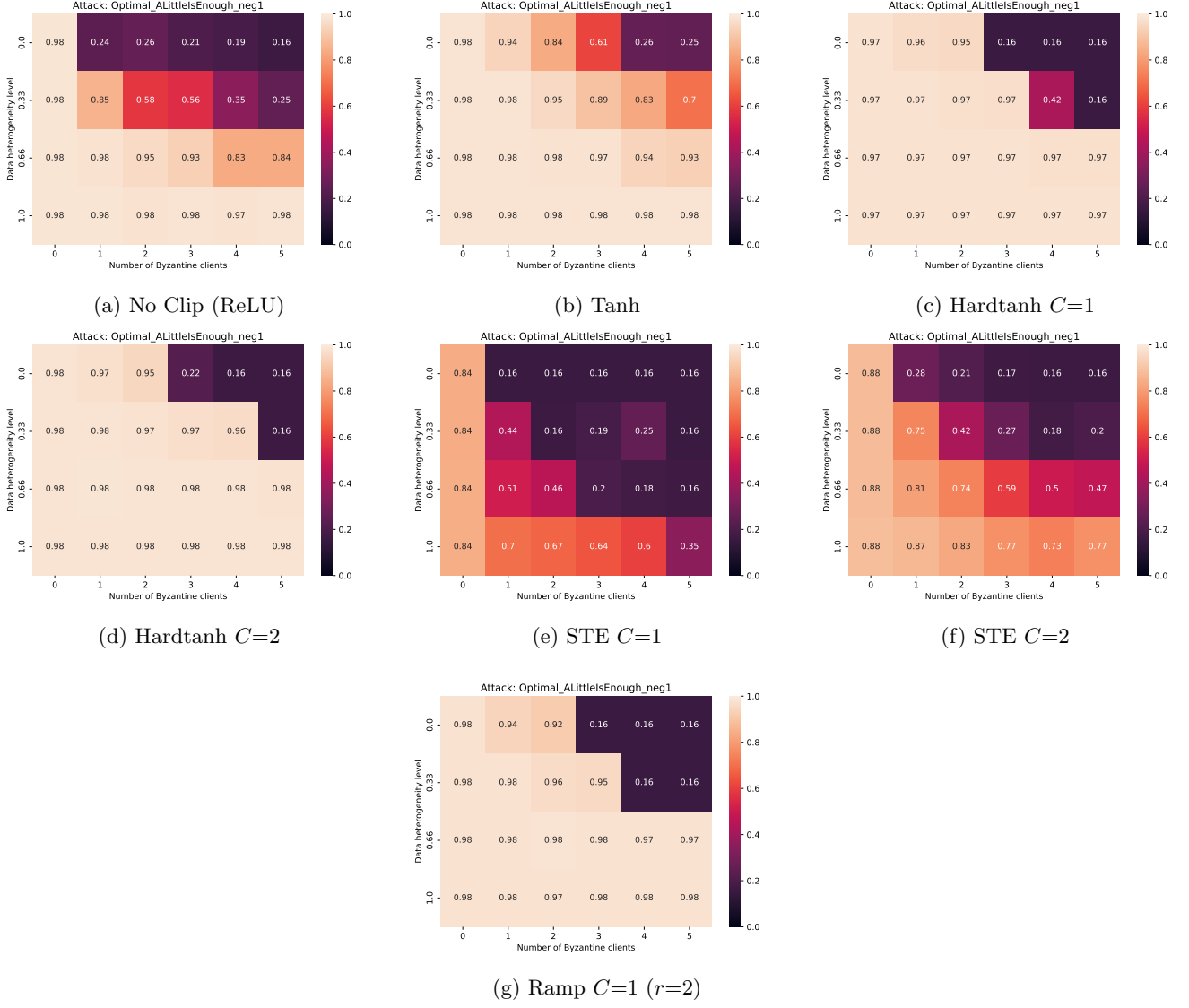


Figure 6: Centered Clipping (CC) under Optimal ALIE Attack.

4.3 CC under SF

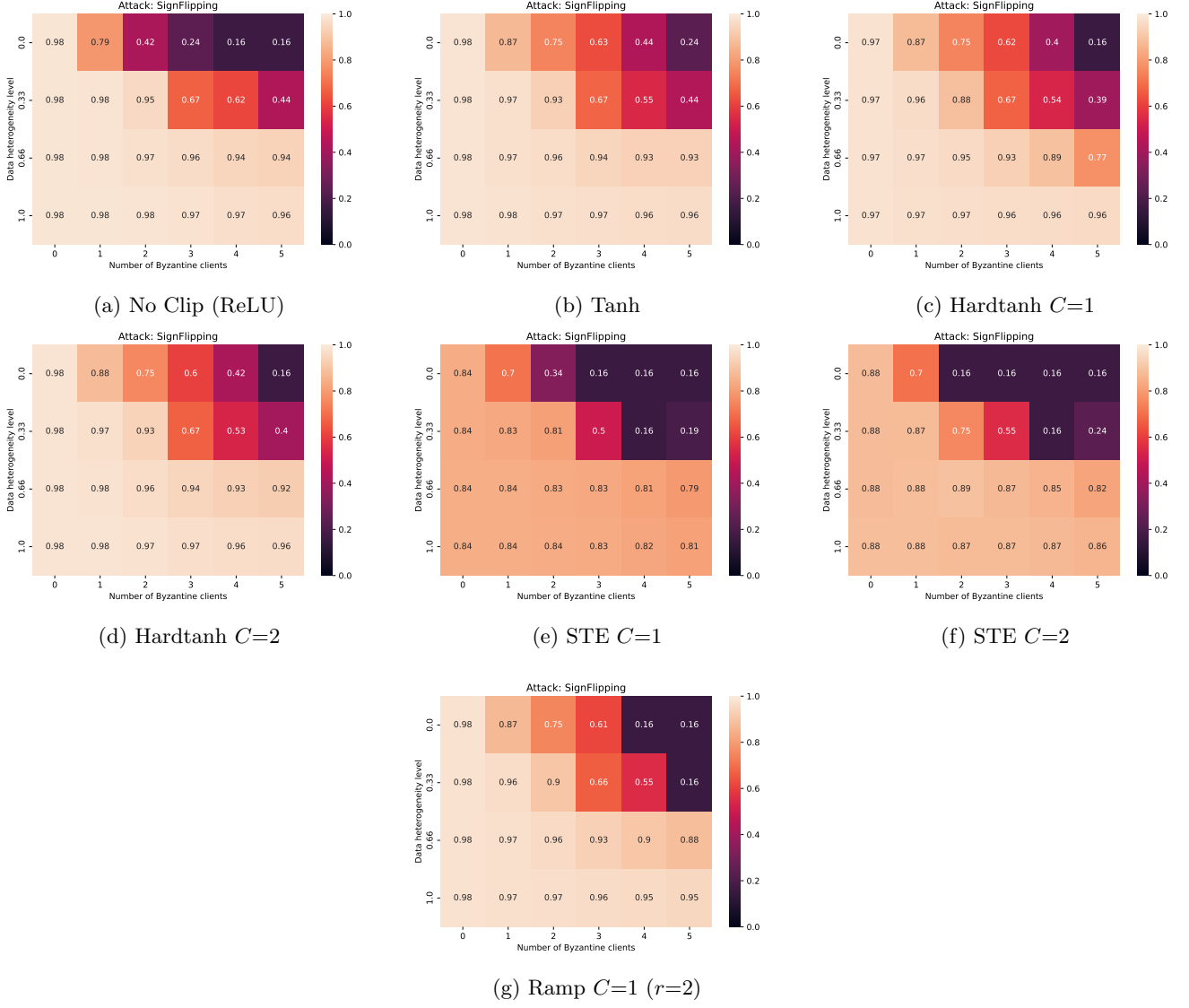


Figure 7: Centered Clipping (CC) under Sign Flipping Attack.

4.4 CC under IPM

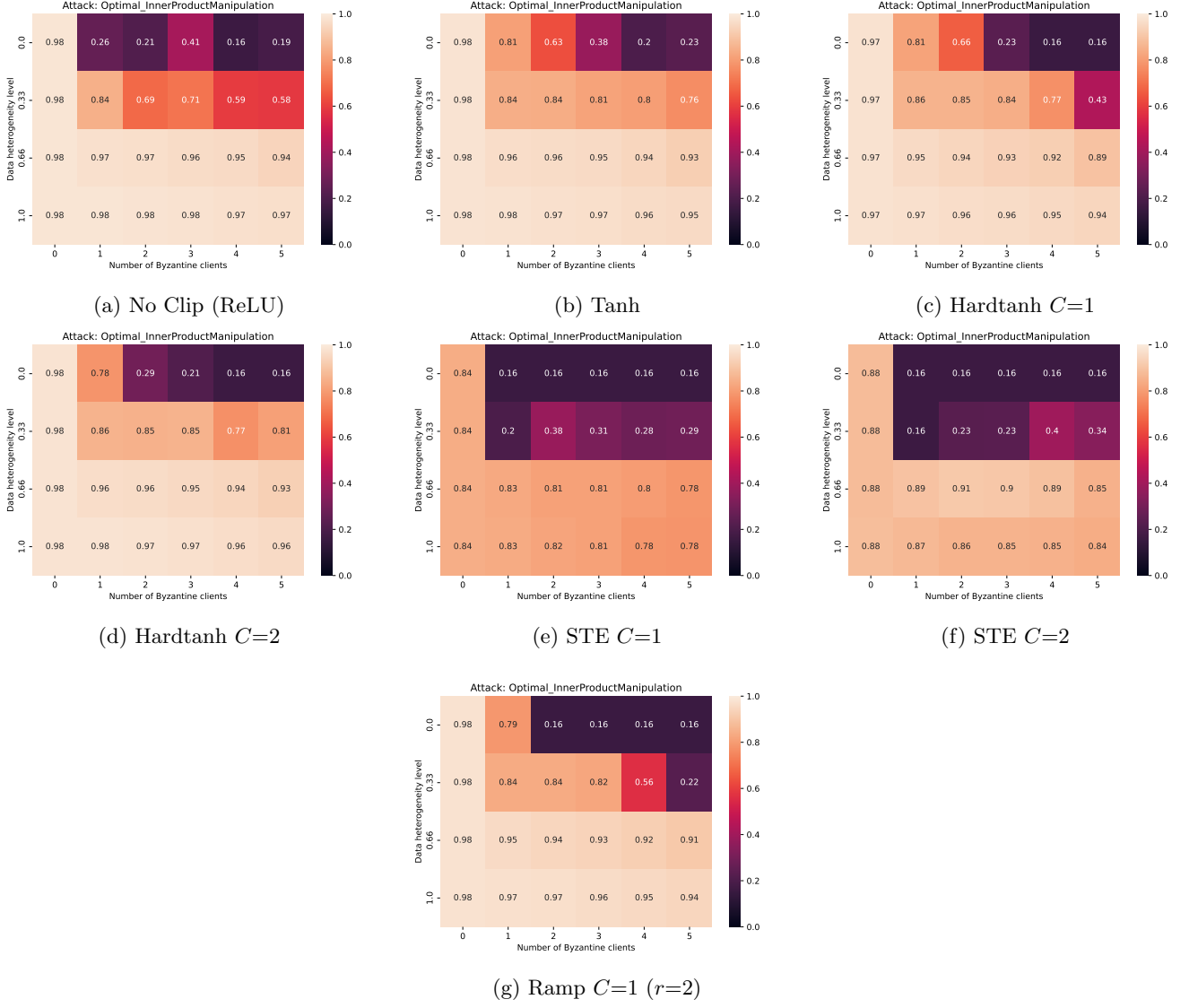


Figure 8: Centered Clipping (CC) under Optimal IPM Attack.

5 Geometric Median (GM)

5.1 GM under Worst-Case across all Attacks

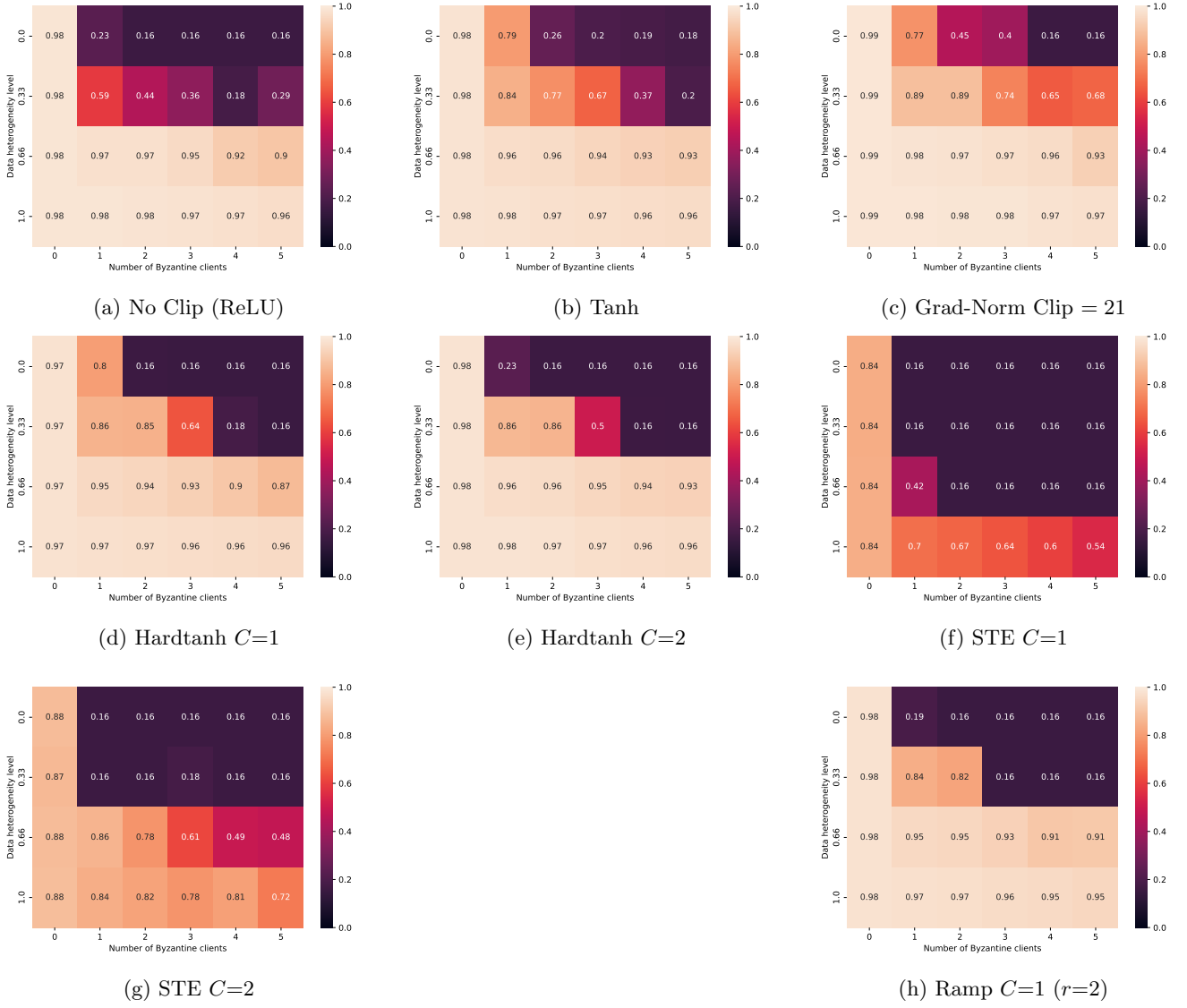


Figure 9: Geometric Median (GM) under Worst-Case across all Attacks.

5.2 GM under ALIE

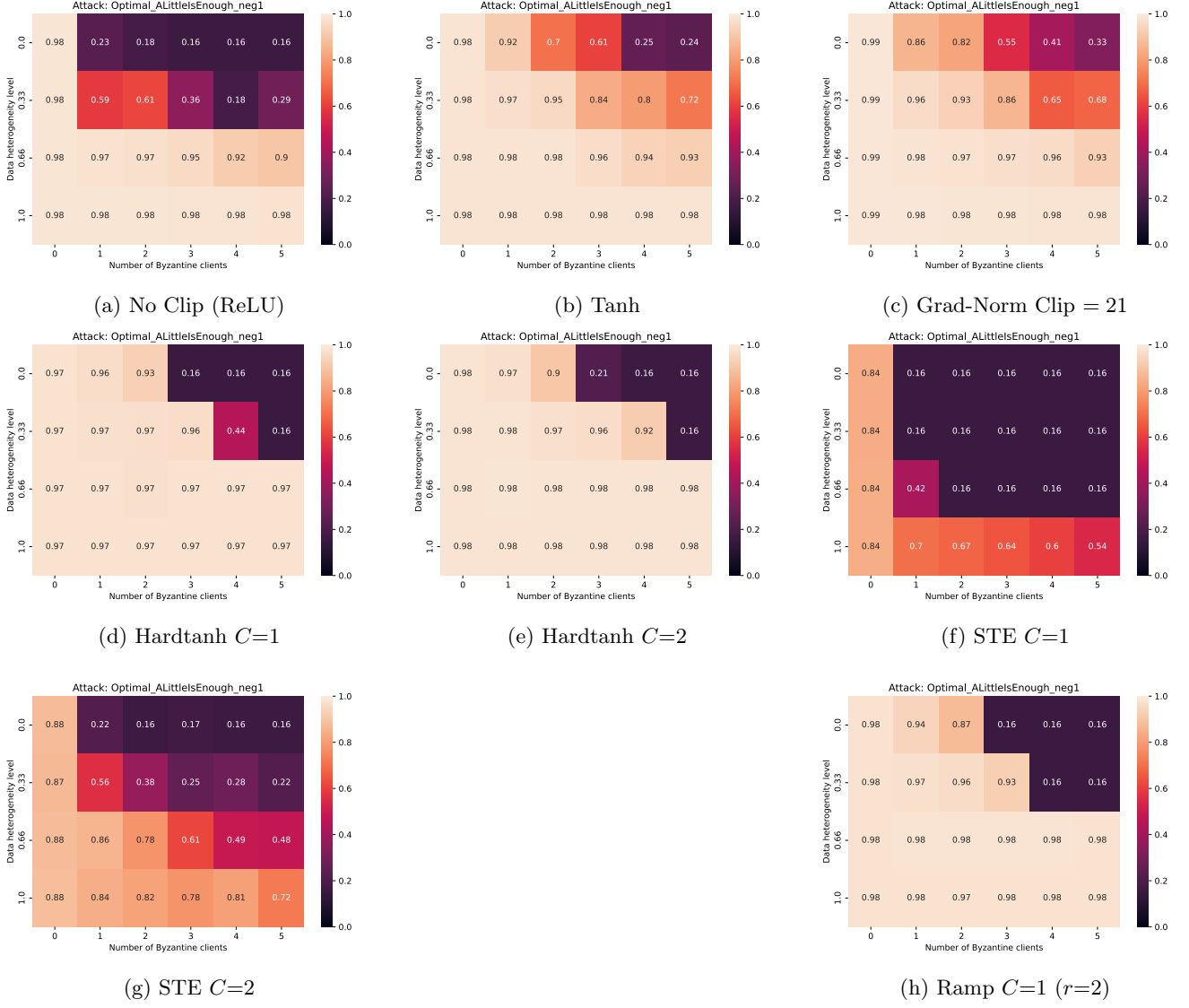


Figure 10: Geometric Median (GM) under Optimal ALIE Attack.

5.3 GM under SF

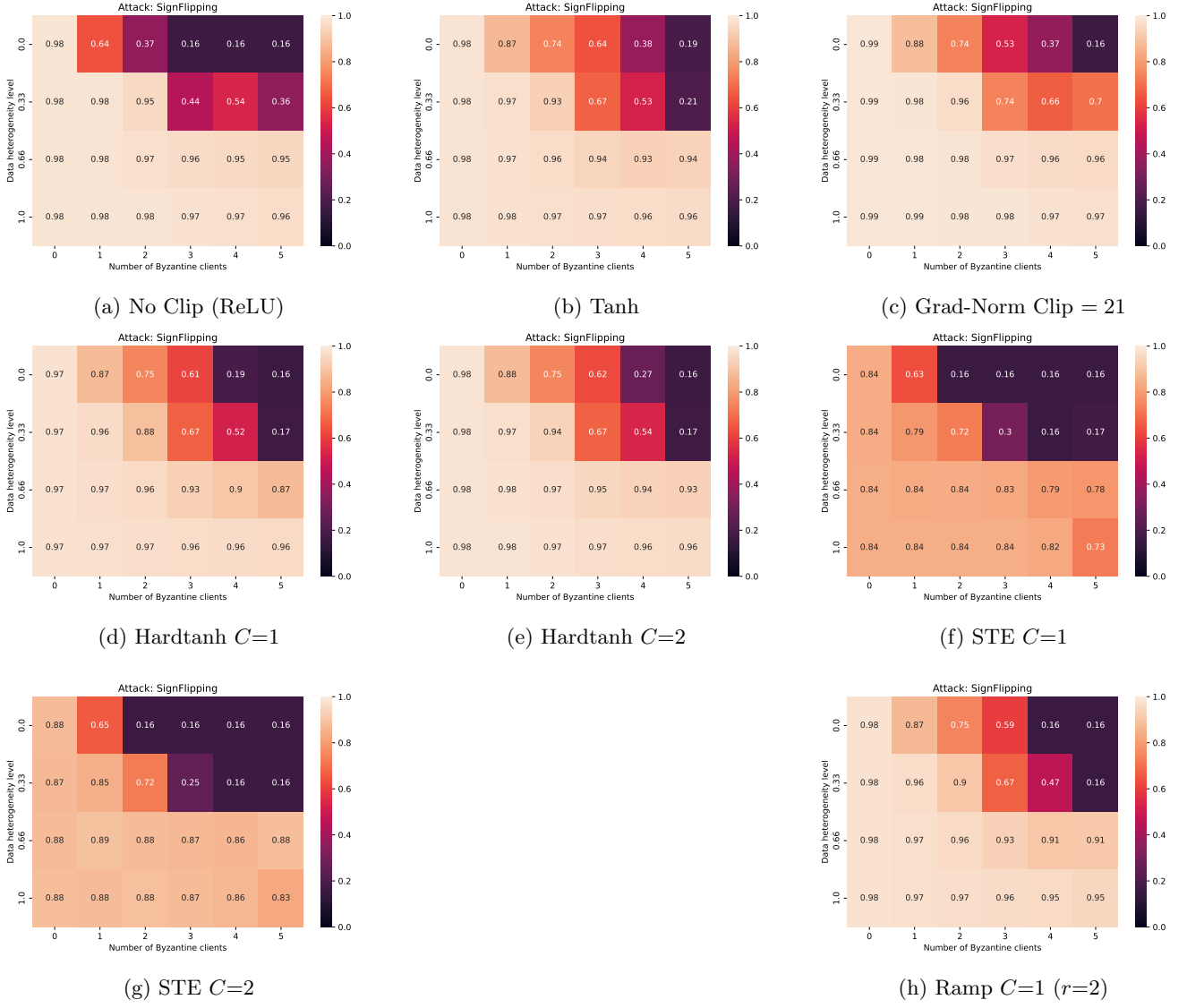


Figure 11: Geometric Median (GM) under Sign Flipping Attack.

5.4 GM under IPM

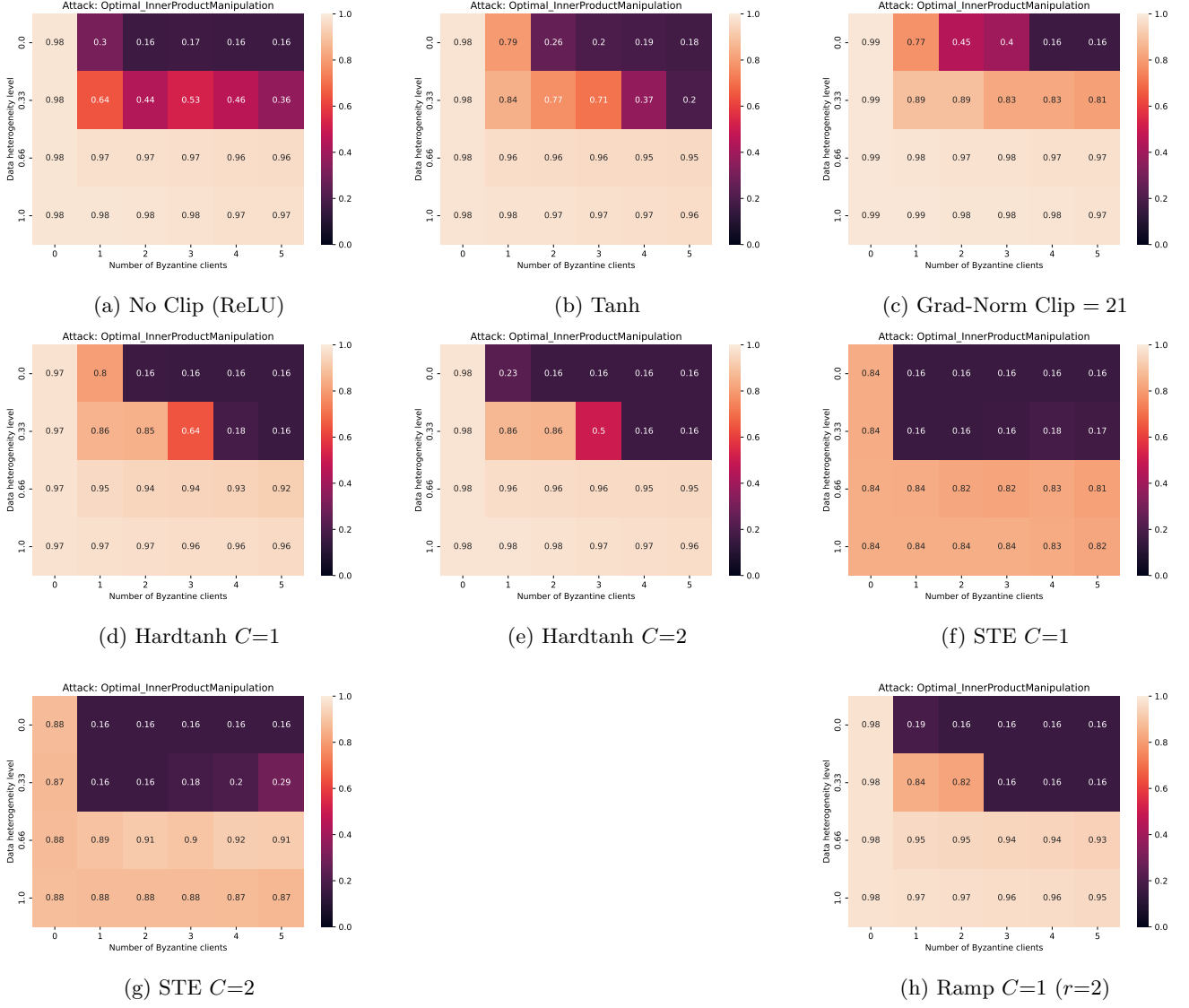


Figure 12: Geometric Median (GM) under Optimal IPM Attack.

6 Multi-Krum (MK)

6.1 MK under Worst-Case across all Attacks

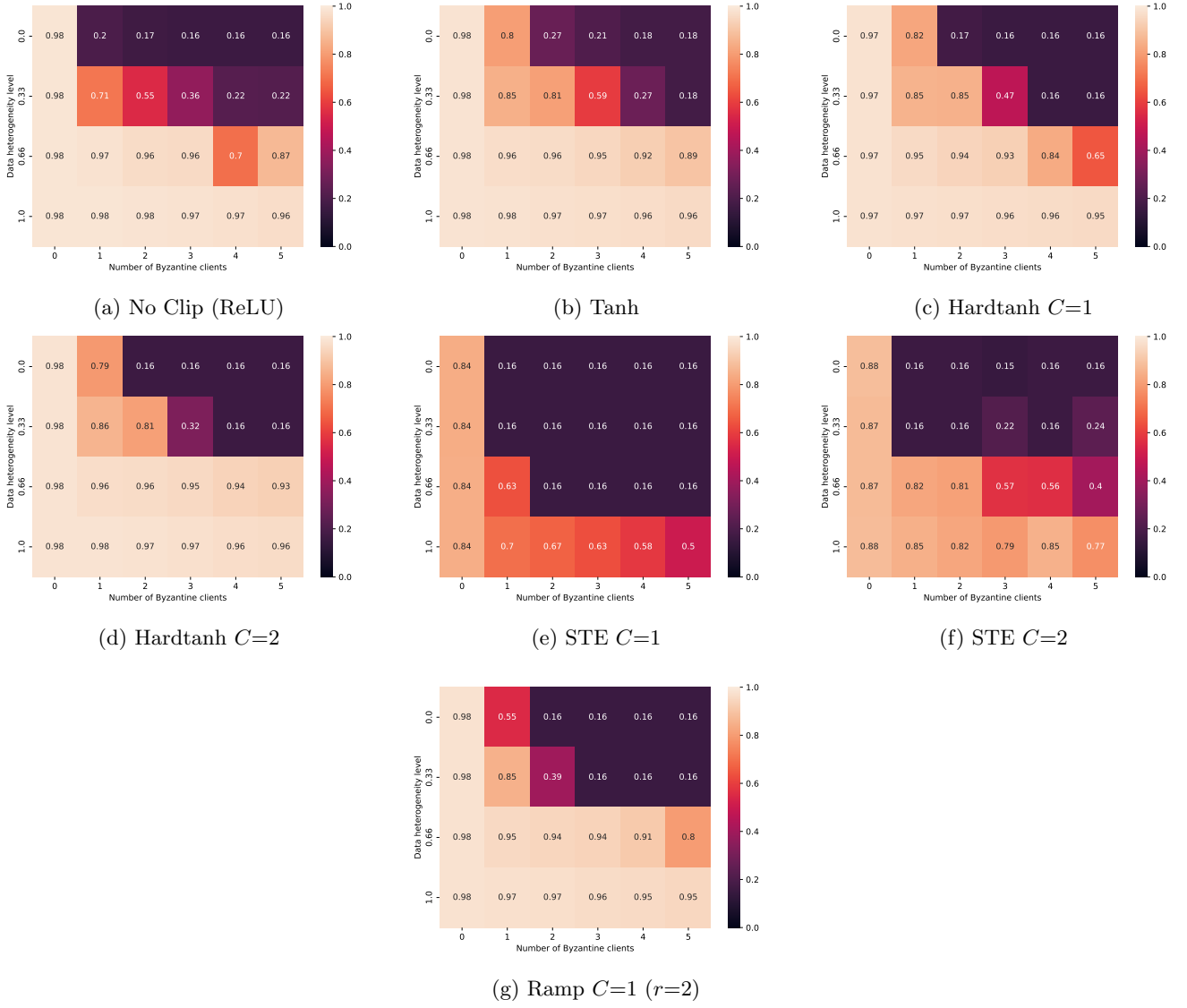


Figure 13: Multi-Krum (MK) under Worst-Case across all Attacks.

6.2 MK under ALIE



Figure 14: Multi-Krum (MK) under Optimal ALIE Attack.

6.3 MK under SF

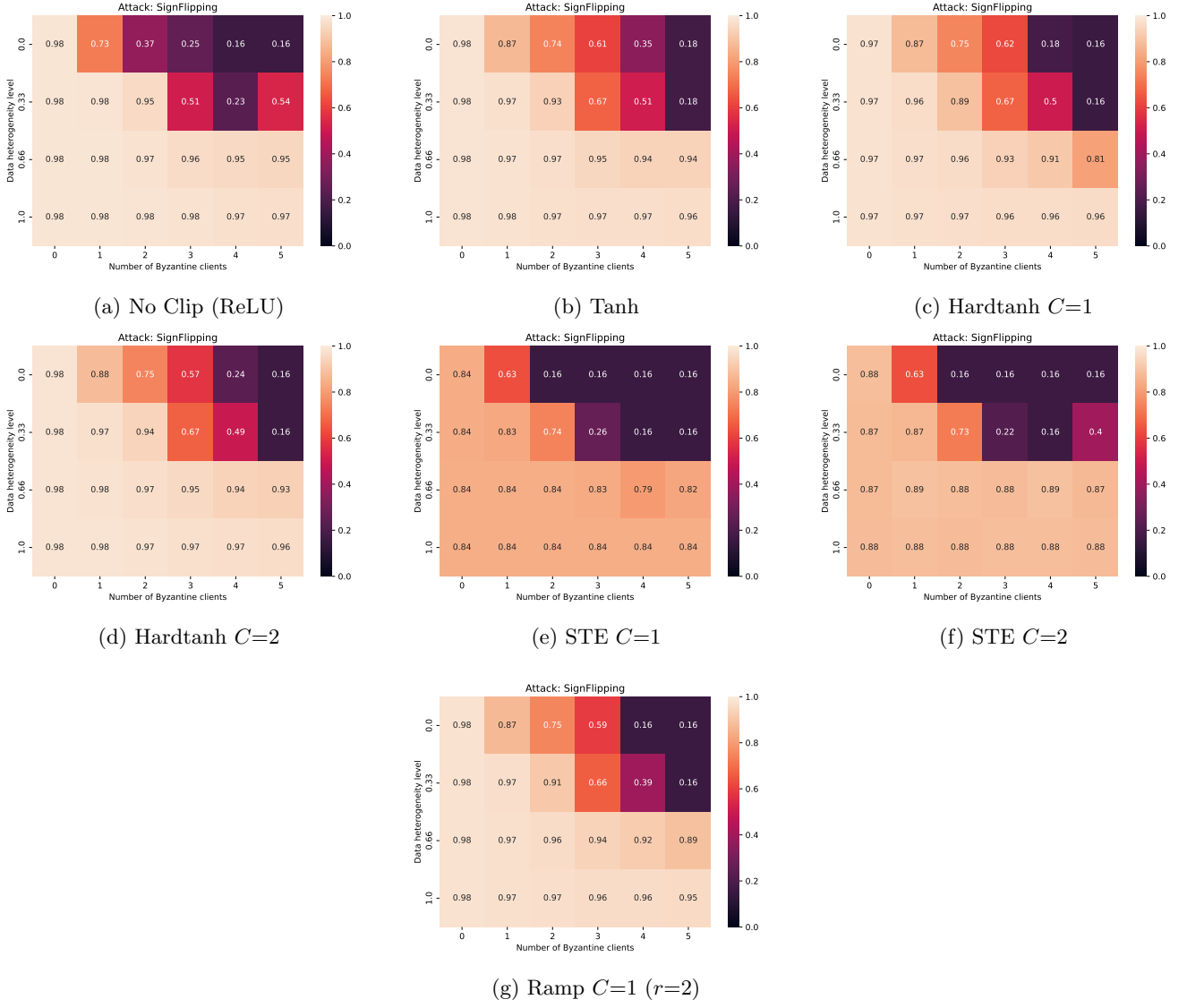


Figure 15: Multi-Krum (MK) under Sign Flipping Attack.

6.4 MK under IPM

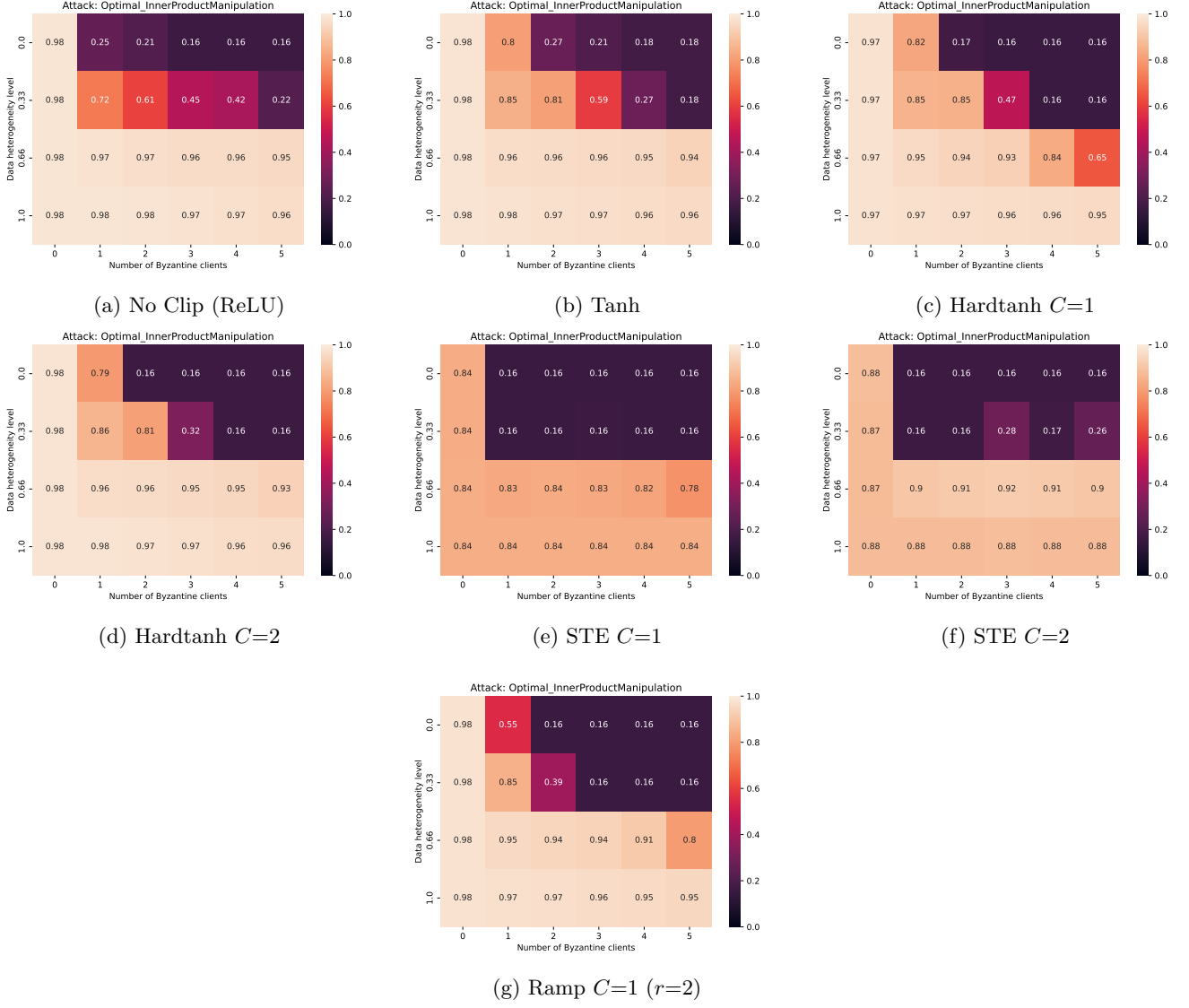


Figure 16: Multi-Krum (MK) under Optimal IPM Attack.

7 Trimmed Mean (TM)

7.1 TM under Worst-Case across all Attacks

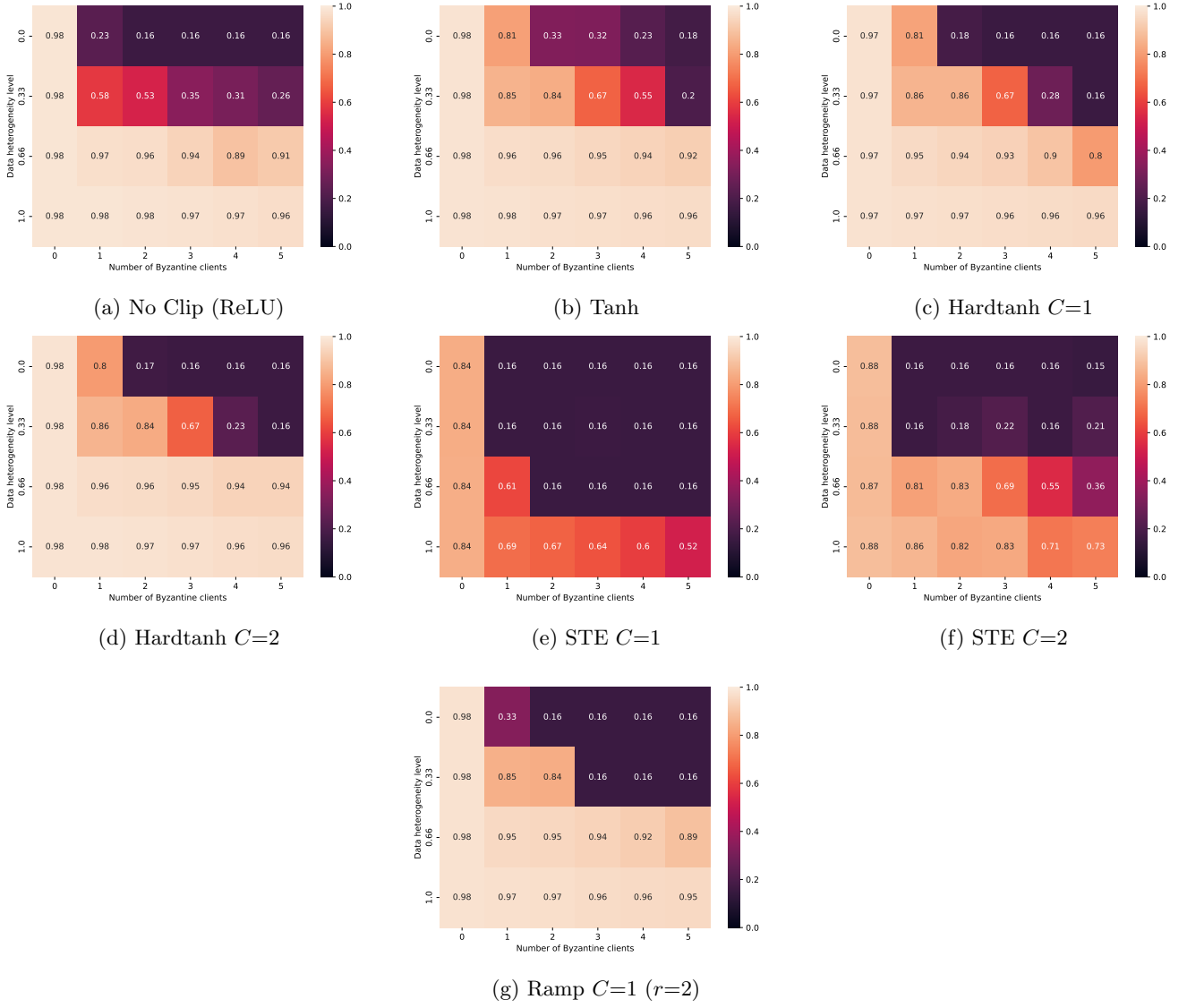


Figure 17: Trimmed Mean (TM) under Worst-Case across all Attacks.

7.2 TM under ALIE



Figure 18: Truncated Mean (TM) under Optimal ALIE Attack.

7.3 TM under SF

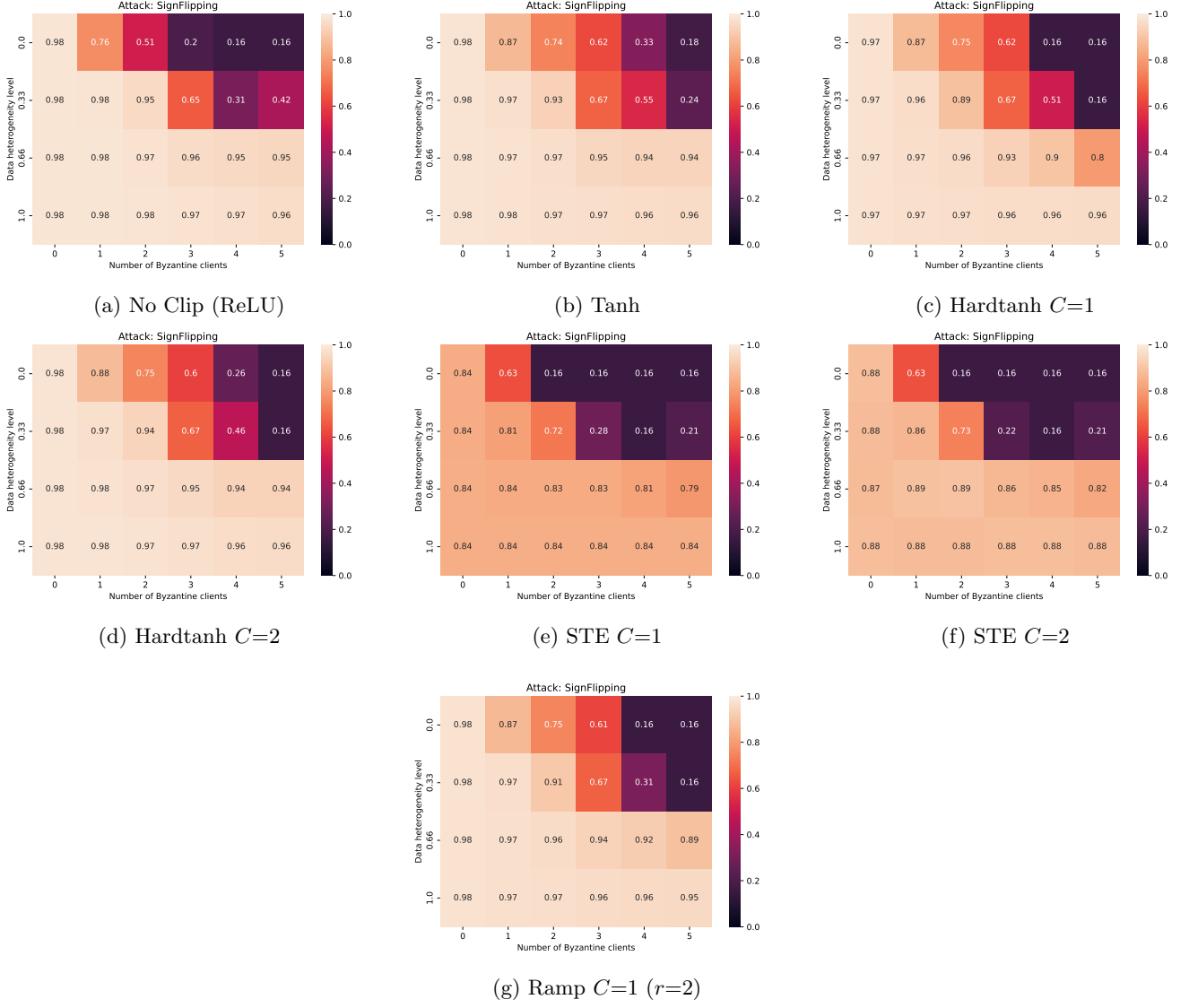


Figure 19: Truncated Mean (TM) under Sign Flipping Attack.

7.4 TM under IPM

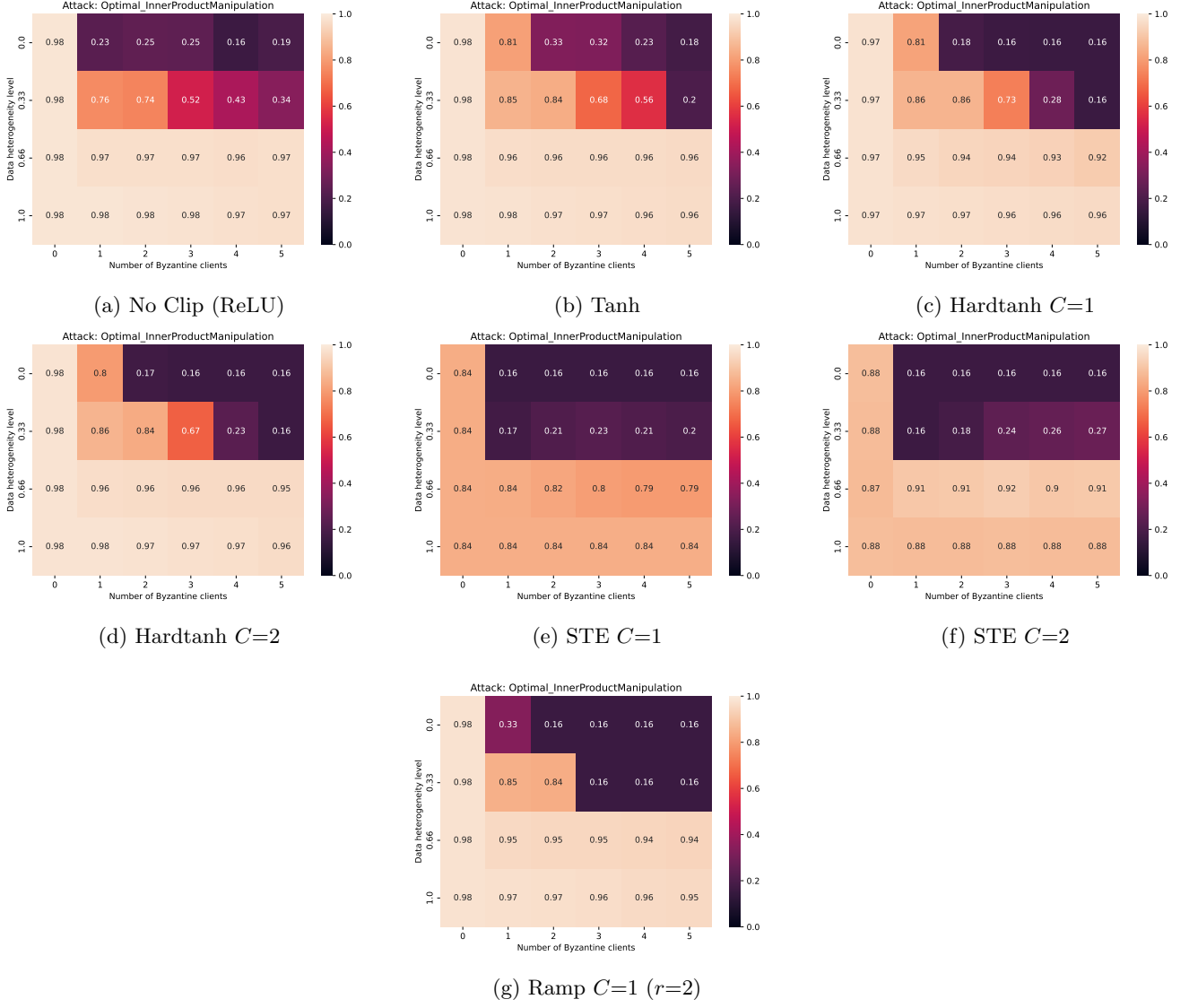


Figure 20: Truncated Mean (TM) under Optimal IPM Attack.